

Selective Insertion with @Transient Annotation

The @Transient annotation in Java is used in JPA/Hibernate to mark a field of a class that should **not** be persisted into the database. Fields marked with @Transient are not included in any insert or update operations in the database. It is useful when you want certain fields to be ignored by JPA, either because they are calculated values or are used for business logic only.

Key Points:

1. Ignore during Persistence:

- Fields marked with @Transient will **not be saved** to the database.
- They will also **not be fetched** from the database.

2. Non-persistent Field:

- These fields are used only in the Java object and **not part of the database schema**.
- JPA ignores the annotated field for insert, update, or query operations.

3. Common Use Cases:

- Fields used for business logic.
- Calculated or derived fields (e.g., age based on dateOfBirth).
- Temporary data, which doesn't need to be stored.

4. Example:

Student.java

```
import jakarta.persistence.Column;
import jakarta.persistence.Entity;
import jakarta.persistence.Id;
import jakarta.persistence.Table;
import jakarta.persistence.Transient;
```

```
@Entity
```

```
@Table(name="StudentTable")
```

```
public class Student {
```

```
    @Id
```

```
@Column(name="sid")  
    private int id;
```

```
@Column(name="sname")  
    private String name;
```

```
@Transient  
    private String city;
```

```
public Student() {  
    System.out.println("Zero param constructor");  
}
```

```
public int getId() {  
    return id;  
}
```

```
public void setId(int id) {  
    this.id = id;  
}
```

```
public String getName() {  
    return name;  
}
```

```
public void setName(String name) {  
    this.name = name;  
}
```

```
public String getCity() {  
    return city;  
}
```

```
public void setCity(String city) {  
    this.city = city;  
}
```

```
}
```

Main.java

Refer Insertion Notes

Output:

```
mysql> use telusko_db;
Database changed
mysql> show tables;
+-----+
| Tables_in_telusko_db |
+-----+
| personinfo           |
| studenttable         |
+-----+
2 rows in set (0.07 sec)

mysql> select * from studenttable
-> ;
+-----+-----+
| sid  | sname |
+-----+-----+
| 13457 | Shiva |
+-----+-----+
1 row in set (0.00 sec)
```

In the above example, the bonus field is annotated with `@Transient` and will not be saved in the database, even though it's part of the Java object.

5. Difference from transient keyword:

- **@Transient:** Specifically for JPA (persistence framework) to ignore the field during database operations.
- **transient keyword:** Used in Java for fields that should not be serialized (during Java object serialization).

6. Important:

- `@Transient` fields can still be used in the application, but they will not be part of the database entity mapping.